Runan He

Curriculum Vitae

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Employment

2024-present **Postdoc**, *Instituto de Ciencias Matemáticas (ICMAT)*, Madrid, Spain.

2021–2024 **Postdoc**, *Martin Luther University Halle-Wittenberg*, Halle(Saale), Germany.

- Mentor: Tomas Dohnal

Education

2017–2021 **Ph.D. in Mathematics**, *Heriot-Watt University, Edinburgh, UK.*

- **Ph.D. Thesis:** "*Analysis of Coupled PDE Systems Modelling Micro-Electro-Mechanical Systems*," under supervision of **Prof. Andrew Lacey, Dr. Heiko Gimperlein,**

2014–2017 **M.Sc. in Pure Mathematics**, Shandong University, Jinan, China.

- **M.Sc. Thesis:** "*Morse Theory and its Application to Semilinear Elliptic Equation*," under supervision of **Prof. Xijun Hu**

2013–2015 **M.Eng. in Software Engineering**, Shandong University, Jinan, China.

- **M.Eng. Thesis:** "*The Design and Implementation of a Reliable Transport System Based on UDP for Large-scale data Transmission*," under supervision of **Prof. Hua Wang**

2009–2013 **B.Sc. in Mathematics and Applied Mathematics**, GPA: 3.0/4, Hunan University, Changsha, China.

- **B.Sc. Thesis:** "*The Existence of Weak Solution to Hypergeometric Equation in two dimensional inhomogeneous Media*," under supervision of **Prof. Lin Yang**

Research Interests

- **Analysis of partial differential equations**
  Well-posedness and singularity formation for nonlinear evolution equations with applications to natural sciences
- **Variational and functional analytic approaches to PDEs**
  Rigorous and formal asymptotic analysis, including calculus of variations, semigroup theory, fixed-point arguments.
- **Modelling in continuum mechanics**
  PDE descriptions of elastic deformation and gas flow, motivated by micro-electro-mechanical systems
- **PDE bifurcation problems**
  bifurcation of nonlinear solutions from spectrum
  bifurcation in PT-symmetric (non-selfadjoint) problems
- **Artificial Intelligence for PDEs**
  Physics-informed neural networks for evolutionary PDEs

Publications and Preprints

- Runan He and Wenjia Xie. Analysis and Numerics of a Stationary Drift-Diffusion Model for Electrical Discharge in MEMS. In preparation.
- Heiko Gimperlein, Runan He and Andrew A. Lacey. Self-similar Solution to Mullins Equation Modelling Sintering. In preparation.
- Heiko Gimperlein, Runan He, Andrew A. Lacey and B. Schörkhuber. Stable quenching for semilinear wave equations. In preparation.
- Heiko Gimperlein, Runan He and Andrew A. Lacey. Analysis of a Model for Electrical Discharge in MEMS. Submitted, arXiv:2602.21439.
- T. Dohnal, M. Hanisch and Runan He. Polychromatic Surface Plasmons in Nonlinear Lossy Media. Submitted, arXiv:2511.21467.
- Tomas Dohnal and Runan He. Bifurcation and asymptotics of cubically nonlinear transverse magnetic surface plasmon polaritons. J. Math. Anal. Appl., 2024, 538(2), https://doi.org/10.1016/j.jmaa.2024.128422.
- Heiko Gimperlein, Runan He and Andrew A. Lacey. Existence and uniqueness for a nonlinear parabolic-hyperbolic coupled system for MEMS. Math. Meth. Appl. Sci., 2024, 1–44, https://doi.org/10.1002/mma.9922.
- Heiko Gimperlein, Runan He and Andrew A. Lacey. Wellposedness of a nonlinear parabolic-dispersive coupled system modelling MEMS. Journal of Differential Equations, 2024, 384: 193-251, https://doi.org/10.1016/j.jde.2023.11.028
- Heiko Gimperlein, Runan He and Andrew A. Lacey. Wellposedness of an elliptic-dispersive coupled system for MEMS. Discrete and Continuous Dynamical Systems, 2023, 43(9): 3485-3511, http://doi.org/10.3934/dcds.2023055
- Heiko Gimperlein, Runan He and Andrew A. Lacey. Quenching for a semi-linear wave equation for micro-electro-mechanical systems. Proceedings of the Royal Society A, 2022, 478(2267): 20220490, https://doi.org/10.1098/rspa.2022.0490
- Runan He. Analysis of coupled PDE systems modelling micro-electro-mechanical systems. PhD thesis, Heriot-Watt University, September 2021, available at: https://www.ros.hw.ac.uk/handle/10399/4629

Conferences and invited seminar talks

07/2026 **Short Communication Speaker**, *International Congress of Mathematicians 2026 (ICM 2026)*, Philadelphia, USA.

- Wellposedness of Nonlinear Coupled PDE Systems Modelling MEMS

10/2025 **Poster Presentation**, *Wave Propagation in Guiding Structures*, Marseille, France.

- Title: Polychromatic Surface Plasmons with Complex Frequencies in Nonlinear Media.

11/2024 **Invited Speaker**, *Research exposition workshop at Instituto de Ciencias Matemáticas (ICMAT)*, Madrid, Spain.

- Wellposedness and singularity formation for PDEs

07/2024 **Invited Speaker**, *16th International Conference on Mathematical and Numerical Aspects of Wave Propagation*, Berlin, Germany.

- Title: Monochromatic and Polychromatic Waves.

03/2024 **Invited Speaker**, *94th Annual Meeting of GAMM*, Magdeburg, Germany.

- Title: Bifurcation and Asymptotics in Nonlinear Surface Plasmon Polaritons.

12/2023 **Invited Speaker**, *Seminar of PDEs group in Technische Universität Clausthal*, Clausthal, Germany.

- Title: Bifurcation and Asymptotics in Maxwell Equations.

07/2022 **Invited Speaker**, *Workshop on Systems Theory and PDEs*, Freiberg, Germany.

- Title: Wellposedness and Singularities in Nonlinear Coupled PDEs Systems for MEMS.

03/2022 **Invited Speaker**, *Conference on Mathematics of Wave Phenomena 2022*, online.

- Title: Analysis of Nonlinear Coupled PDEs Systems for MEMS.

10/2021 **Invited Speaker**, *Oberseminar at Martin-Luther University Halle-Wittenberg*, Halle (Saale), Germany.

- Title: Analysis for MEMS models

06/2020 **Invited Speaker**, *Maxwell Institute Postgraduate Seminar*, Edinburgh, United Kingdom.

- Title: Well-posedness of a Nonlinear Elliptic-Dispersive Coupled System for MEMS.

06/2019 **Invited Speaker**, *Maxwell Institute Postgraduate Seminar*, Edinburgh, United Kingdom.

- Title: Finite-time Existence for a Nonlinear Parabolic-Dispersive Coupled System for MEMS with an Elastic Plate.

11/2018 **Invited Speaker**, *Heriot-Watt Postgraduate Seminar*, Edinburgh, United Kingdom.

- Title: Operator Semigroup Theory and its Application to PDEs.

06/2018 **Invited Speaker**, *Heriot-Watt Applied Mathematics Postgraduate Seminar*, Edinburgh, United Kingdom.

- Title: Well-posedness for a Nonlinear Parabolic-Hyperbolic Coupled System for MEMS.

Teaching Experience and Service

2021–2024 **Lecturer at Martin Luther University Halle-Wittenberg**, *Halle(Saale)*, Germany.

- Lecture Courses: Evolution Equation, Fachseminar.
- Exercise Courses: Partial Differential Equations, Dynamical System, Mathematical Physics (in German), Nonlinear Analysis.

2017–2021 **Tutorial Assistant at Heriot-Watt University**, *Edinburgh*, United Kingdom.

- Tutor courses: Calculus, Mathematics for Scientist and Engineer, PDE, Numerical Analysis, Complex Analysis, Functional Analysis, Real Analysis, Mathematical Experiment.

2018–2019 **Assistant at public lectures of Heriot-Watt University**, *Edinburgh*, United Kingdom.

- Assist the public lectures "Masterclass" for Edinburgh High School Students.

2014–2017 **Teaching and Research Assistant at Shandong University**, *Jinan*, China.

- Assist the undergraduate courses at Shandong University
- Organize seminars on Calculus of Variations and PDEs.

2013–2014 **Computer Programmer at Shandong University**, *Jinan*, China.

- Create and manage the schedule of software development.

Research Projects

2024–present **Project title: Dynamics of singular incompressible fluids**, *Funding source: Ministerio de Ciencia e Innovacion, PID2023-152878NB-I00*, Principle investigator: Diego Córdoba (ICMAT), Period: September 2024 – August 2027.
Amount: 70.000 Euros

Personal contribution: In this project, I am working as a researcher, pursuing new methods in the mathematical analysis of nonlocal and nonlinear partial differential equations (PDEs). I am also investigating the mathematical models describing the incompressible fluids and their interaction with

some elastic structure (Fluid-Structure interaction).

2024–present **Project title: ICMAT Severo Ochoa project**, *Funding source: Ministerio de Ciencia e Innovacion, CEX2023-001347-S*, Principle investigator: Diego Córdoba (ICMAT), Period: June 2024 – May 2028.
Amount: 5.660.000 Euros

Personal contribution: In this project, I am working as a researcher, modelling and analysing the various fluid dynamics

Awards and Honors

- 2015 Third Ranking Graduate Academic Scholarship, Shandong University, China.
- 2014 Exceptional Graduate Academic Scholarship, Shandong University, China.
- 2012 Third Prize in Mathematics Competition in Hunan Province, China.
- 2010 Second Prize in Mathematics Competition in Hunan Province, China.

References

- **Prof. Tomas Dohnal**
  Martin-Luther-Universität Halle-Wittenberg,
  Halle(Saale), Germany
  tomas.dohnal@mathematik.uni-halle.de
- **Prof. Andrew Lacey**
  Heriot-Watt University
  Edinburgh, UK
  a.a.lacey@hw.ac.uk
- **Prof. Heiko Gimperlein**
  Leopold-Franzens-Universität Innsbruck
  Innsbruck, Austria
  Heiko.Gimperlein@uibk.ac.at
- **Prof. Dominic Breit**
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  Clausthal-Zellerfeld, Germany
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